



BRAC UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in CS / CSE Program
Mid-Term Exam, Fall 2023

Course: CSE437 (Data Science: Working with Real World Data)
Full Marks: 25
Time: 90 minutes

Note: Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

1. **Apply** Gradient Descent to find out the optimal values of the learning parameters for predicting the GPA using the following dataset. You have to show 1 iteration only. Use mean squared error ($MSE = \frac{\sum_{i=1}^n ((y_i - \hat{y}_i)^2)}{n}$) as the loss function and consider the learning rate as 0.01 and encode the categorical variables as Good=2 and Poor=1. [CO2,C3, Mark: 8 + 2]

Observations	Hours of Study	Soundness in Mathematics	Soundness in English	GPA
1	2	Good	Poor	3
2	4	Good	Good	4
3	6	Poor	Poor	2
4	8	Poor	Good	2.5
5	10	Good	Poor	3
6	12	Poor	Good	4

Afterward, **predict** GPA for the following query using the optimal values of the learning parameters.

Hours of Study	Soundness in Mathematics	Soundness in English
5	Good	Poor

2. When you're looking for a community to live in, there are a lot of factors you might consider, e.g., home affordability, proximity to work, and quality of the nearby natural environment. Livability is a measurement of how attractive a neighborhood, city, and/or region is for you based on a variety of factors. [CO2,C2, Mark: 8 + 2]
Form 2 clusters using the K-Means clustering algorithm to group similar livable cities together; considering the *Safeness score*, *Environment*, and *Affordability* features of the following dataset. Use Euclidian distance to measure the dissimilarity among the cities. Encode the categorical variables as Excellent=3, Good=2, Poor=1 and Expensive = 2, Affordable=1.

No.	City	Safeness score	Environment	Affordability
1	Tokyo	88	Excellent	Expensive
2	Seoul	86	Good	Affordable
3	Singapore	84	Good	Expensive

4	Delhi	50	Unhealthy	Affordable
5	New York	70	Good	Expensive
6	Karachi	40	Unhealthy	Affordable
7	Amsterdam	90	Excellent	Affordable

Afterward, **determine** the cluster of Dhaka city from the formed cluster based on the following feature values:

<i>City</i>	<i>Safeness score</i>	<i>Environment</i>	<i>Affordability</i>	<i>Cluster?</i>
Dhaka	52	Unhealthy	Expensive	?

3. What are the limitations of a decision tree? How to overcome those limitations? How Adaboost is different from a decision classifier? How to assign the weight of the vote of each stump of the Adaboost classifier? How does Adaboost ensure more attention to the misclassified instances?

[CO1,C1,
Mark: 5]